

Standard Form & Indices

Test 6 · 41 marks total · Sections F & G include Higher-tier content

Name: _____

Date: _____

Score: _____ / 41

Section A — Laws of indices

A1. Write $7^4 \times 7^5$ as a single power of 7.

[1 mark]

A2. Write $5^{12} \div 5^7$ as a single power of 5.

[1 mark]

A3. Write $(3^4)^2$ as a single power of 3.

[1 mark]

A4. Work out the value of $2^4 \times 2^2$.

[2 marks]

Section B — Zero and negative indices

B1. Work out the value of 100^0 .

[1 mark]

B2. Work out the value of 7^{-2} . Give your answer as a fraction.

[2 marks]

B3. Work out the value of 10^{-7} . Give your answer as a decimal.

[2 marks]

Section C — Writing standard form

C1. Write 36,000 in standard form.

[2 marks]

C2. Write 7,400,000 in standard form.

[2 marks]

C3. Write 0.0052 in standard form.

[2 marks]

C4. Write 0.0007 in standard form.

[2 marks]

Section D — Converting back, and calculating

D1. Write 8.3×10^3 as an ordinary number.

[1 mark]

D2. Write 2×10^{-6} as an ordinary number.

[1 mark]

D3. Work out $(7 \times 10^3) \times (3 \times 10^5)$. Give your answer in standard form.

[3 marks]

D4. Work out $(3 \times 10^9) \div (6 \times 10^4)$. Give your answer in standard form.

[3 marks]

Section E — Powers to know, and estimating roots

E1. Write down the value of 2^8 .

[1 mark]

E2. $\sqrt{50}$ lies between which two consecutive whole numbers?

[1 mark]

Section F — Laws of indices with algebra

F1. Simplify $x^9 \times x^3$.

[1 mark]

F2. Simplify $y^{14} \div y^9$.

[1 mark]

F3. Simplify $(z^6)^2$.

[1 mark]

F4. Simplify $3m^5 \times 8m^2$.

[2 marks]

F5. Simplify $(4n^2)^3$.

[2 marks]

Section G — Fractional indices (Higher)

G1. Work out the value of $100^{1/2}$.

[1 mark]

G2. Work out the value of $216^{1/3}$.

[1 mark]

G3. Work out the value of $32^{3/5}$.

[2 marks]

G4. Work out the value of $36^{-1/2}$.

[2 marks]